

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (IT)

April 2013

IT206 Object Oriented Programming using Java

Date: 23.04.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION - I

- Q - 1 (a) List and explain the features of Java programming language. [04]  
(b) What is data type? Explain type casting with suitable example. [03]
- Q - 2 (a) What are The Scope and Lifetime of Variables? Write the differences between class variable and local variable. [04]  
(b) What are access specifiers? List all the access specifiers available in Java. [03]  
(c) Write a program to read five numbers as a command line argument and sort the numbers in ascending order. [07]

## OR

- Q - 2 (a) Explain the Bitwise Logical Operators with suitable example. [03]  
(b) Explain the importance of 'this' reference. Write a code to depicts the use of 'this' reference [04]  
(c) Write a program to display given numbers (1 to 100) in words. [07]
- Q - 3 (a) Write the differences between abstract class and interface. [03]  
(b) By taking suitable explain method overloading and constructor overloading [04]  
(c) Explain static variables and static methods in Java. Provide an example to explain them. [07]

## OR

- Q - 3 (a) What are Short-Circuit Logical Operators? How does it give performance improvement over normal bitwise operator? [03]  
(b) Explain Java Garbage collector. [04]  
(c) How final access modifier affect in declaration of class, method and variable? How constant is declared in Java? [07]

## SECTION – II

- |                  |                                                                                                                                                                      |             |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Q - 4 (a)</b> | Explain how to use thread class for Multithreading in Java. Explain with an example.                                                                                 | <b>[05]</b> |
| <b>(b)</b>       | Which are the streams available in Java for I/O? What is buffered stream?                                                                                            | <b>[02]</b> |
| <b>Q - 5 (a)</b> | What is Layout manager? Explain different layout manager in Java.                                                                                                    | <b>[07]</b> |
| <b>(b)</b>       | Discuss the overview of Object Oriented Analysis. Describe the basic activities in Object Oriented Analysis and explain how Use-case Modeling is useful in analysis. | <b>[07]</b> |

OR

- Q - 5 (a)** Why we need object oriented programming techniques? Give a brief note on object relationships and association with an example. [07]
- (b)** Explain the Exceptions categories, i.e. checked and unchecked exceptions. Explain try, catch, finally, throw and throws keywords. [07]
- Q - 6 (a)** Define the Generalization notation in UML class diagram. How it is useful in Object oriented design. [08]
- (b)** When to use the following diagram: [06]
- a. Package diagram
  - b. Deployment diagram
  - c. Interaction diagram

Take suitable example for explanation.

OR

- Q - 6 (a)** Draw Interaction diagram (either sequence or Collaboration) for money transaction process of ATM system. **[08]**
- (b) Explain the following terms (Any four):** **[06]**
- Swimlanes
  - Dynamic Concurrency
  - Realization
  - Multiple Classification
  - Dynamic Classification
  - Composition

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